Math 133: Derivatives and Tangent Lines

Name: _____ October 14, 2026

1. Warmup. Compute each derivative using the power, sum, difference, and constant multiple rules. Rewrite roots and reciprocals as powers when useful. Show your work.

(a) $p(x) = 2x^4 - 5x^2 + 6$

(b) $r(\theta) = \frac{1}{2}\theta^6 - 4\theta + 8$

(c) $s(x) = \dfrac{8 + x}{x^2}, \quad x \neq 0$

(d) $u(z) = 3\sqrt{z} + 2z, \quad z > 0$

Product rule: If f, g are smooth functions, then $(fg)' =$ _____.

(e) $\varphi(x) = \sqrt{1 + 2x}, \quad x > -1/2$

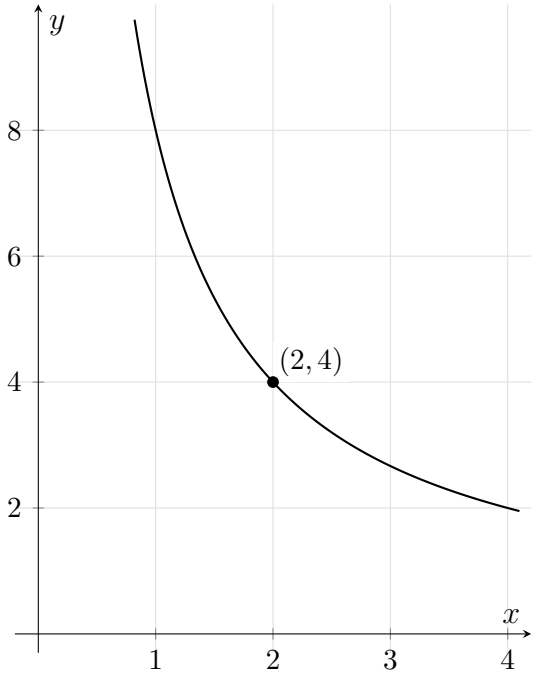
(f) $\sqrt{1 + 2x} \cdot x^2, \quad x > -1/2.$

2. Let $g(x) = \dfrac{8}{x}$ for $x > 0$. Its graph and the point $(2, 4)$ are shown.

(a) Find $g'(x)$ and the slope at $x = 2$.

(b) Find an equation $y = L(x)$ of the tangent line at $(2, 4)$. Sketch the line on the graph.

(c) Use $L(x)$ to estimate $g(1.9)$ and $g(2.1)$. Show each substitution.



3. A company produces insulated bottles. Let $C(q)$ be the total cost, in dollars, of producing q bottles. No formula for C is available, but

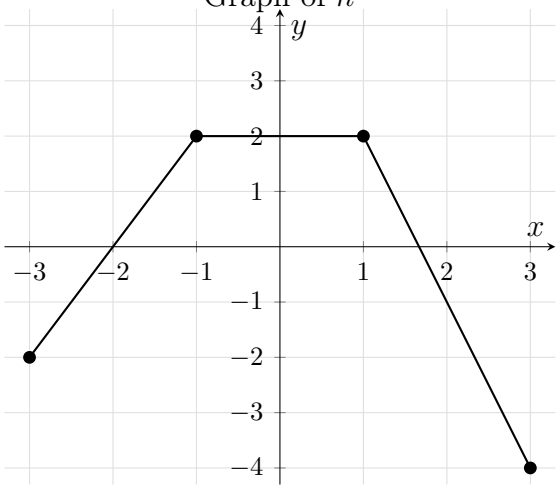
$$C(200) = 4800, \qquad C'(200) = 17.$$

Estimate $C(205)$ and $C(198)$ using this information. Show your work.

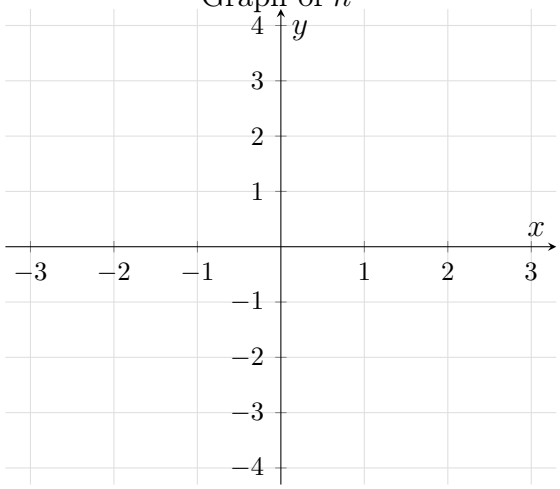
Hint: What is the tangent line approximation for $C(q)$ near $q = 200$?

4. The graph of a piecewise linear function h is shown below. Sketch the graph of h' for $-3 < x < 3$, and indicate where the derivative does not exist.

Graph of h



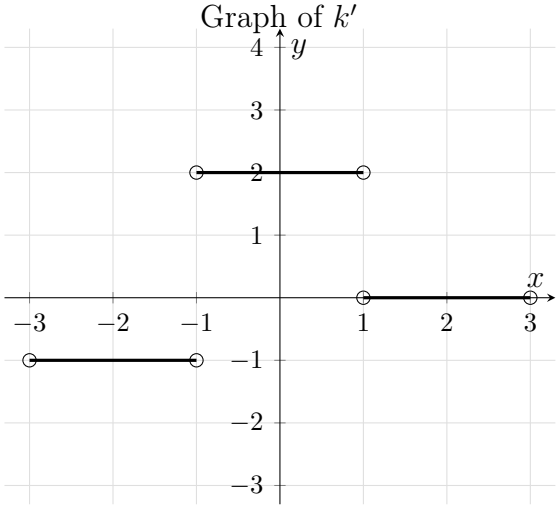
Graph of h'



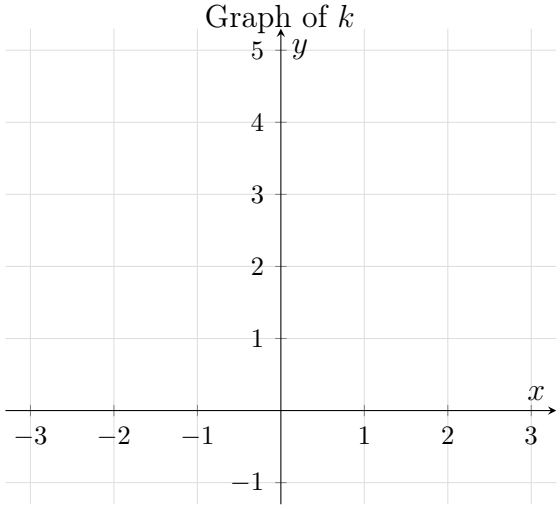
h' does not exist at $x =$ _____.

5. A continuous function k on $[-3, 3]$ has the derivative shown below wherever it exists. Given $k(-3) = 2$, sketch the graph of k.

Graph of k'

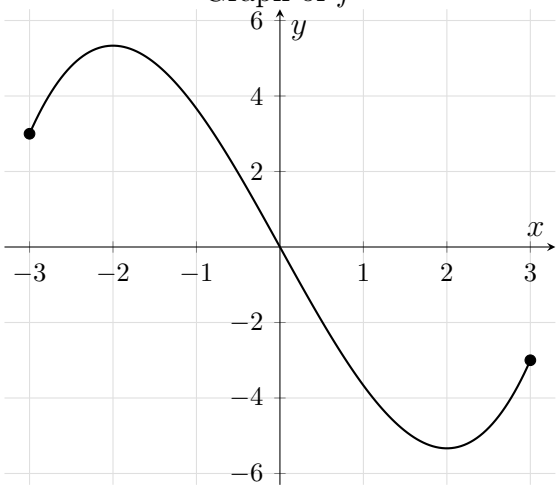


Graph of k

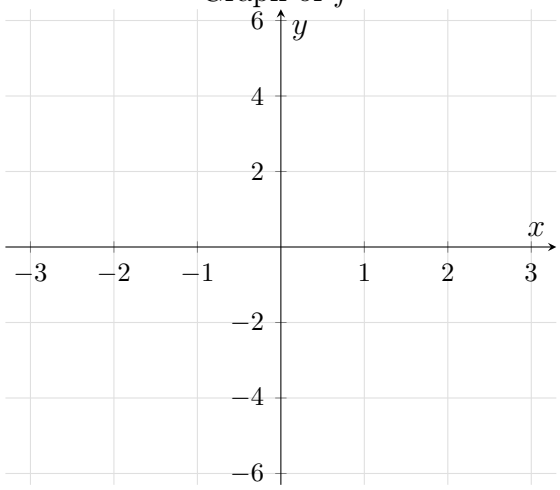


6. The graph of a smooth function f is shown below. Sketch f' on the blank axes, and complete the statements below using the graphs.

Graph of f



Graph of f'



For $x \in [-3, 3]$:

$f(x) > 0$ when $x \in$ _____

$f(x) = 0$ when $x =$ _____

$f(x) < 0$ when $x \in$ _____

For $-3 < x < 3$:

$f'(x) > 0$ when $x \in$ _____

$f'(x) = 0$ when $x =$ _____

$f'(x) < 0$ when $x \in$ _____

f is increasing on: _____

f is decreasing on: _____

On $[-3, 3]$, estimate the maximum and minimum values of f, and give where each occurs.

Maximum value: _____, occurring at $x =$ _____.

Minimum value: _____, occurring at $x =$ _____.

Notes

Let f be a smooth function on an interval containing a.

If $f'(a) > 0$, then f _____ at a.

If $f'(a) < 0$, then f _____ at a.

If $f'(a) = 0$, then f _____ at a.